6-WEEK INTENSIVE PROGRAM

# PYTHON FOR DATA SCIENCE

Master Python from the ground up — data structures, OOP, file handling, and everything you need to start your Data Science journey.

| 6 Weeks | 8 Modules | 2 Projects | Live Sessions | 100% Support |
|---|---|---|---|---|
| Duration | Curriculum | Hands-On | Expert-Led | Placement |

## 6-WEEK LEARNING ROADMAP

A perfectly structured week-by-week journey — from zero to writing real Python programs.

| W1 | W2 | W3 | W4 | W5 | W6 |
|---|---|---|---|---|---|
| Python Basics | Control Flow | Functions & OOP | Data Structures | File I/O & Modules | Projects & Review |

## COMPLETE CURRICULUM BREAKDOWN

Every concept, every skill — in precise detail. No gaps, no shortcuts.

MOD 01

### Python Setup & Fundamentals

- Installing Python, Anaconda & VS Code
- Understanding the Python interpreter
- Variables & naming conventions
- Data Types: int, float, complex
- Strings: creation, slicing, methods
- Booleans & None type
- Type Casting & type() function
- Arithmetic & Assignment operators
- Comparison & Logical operators
- Taking user input & print formatting

MOD 02

### Control Flow & Decision Making

- if / elif / else statements
- Nested conditionals
- Comparison chaining
- while loops — basic & advanced
- for loops — range & iterables
- break, continue & pass
- Nested loops
- Loop else clause
- Ternary (one-line) expressions
- Pattern printing exercises

MOD 03

## Functions & Scope

- Defining & calling functions
- Parameters vs Arguments
- Default & keyword arguments
- *args and **kwargs
- Return values & multiple returns
- Variable scope: local vs global
- Lambda (anonymous) functions
- map(), filter() & reduce()
- Recursion & base cases
- Higher-order functions

MOD 04

## Data Structures — Lists & Tuples

- Creating & accessing Lists
- Indexing & negative indexing
- Slicing lists
- List methods: append, insert, pop, sort
- List comprehensions
- Nested lists & 2D lists
- Tuples — immutability & use cases
- Tuple packing & unpacking
- Zipping & enumerating
- Sorting with key functions

MOD 05

## Data Structures — Sets & Dictionaries

- Sets — creation & properties
- Set operations: union, intersection, diff
- Frozensets
- Dictionaries — key-value pairs
- Dict methods: get, keys, values, items
- Nested dictionaries
- Dictionary comprehensions
- Counting & grouping with dicts
- defaultdict & Counter (collections)
- Choosing the right data structure

MOD 06

## Object-Oriented Programming (OOP)

- Classes & Objects — concepts
- Defining a class & __init__ method
- Instance variables & methods
- Class variables & class methods
- Static methods
- Inheritance & method overriding
- super() keyword
- Multiple inheritance
- Encapsulation & name mangling
- Polymorphism & duck typing

MOD 07

## Exception Handling, File I/O & Modules

- try / except / else / finally
- Catching specific exceptions
- Raising custom exceptions
- Reading & writing text files
- Working with CSV files
- JSON: read, write & parse
- Context managers (with statement)
- Built-in modules: os, sys, math, random
- Creating your own modules & packages
- Using pip to install libraries

MOD
08

### Iterators, Generators & Modern Python

- Iterators & the iterator protocol
- Creating custom iterators
- Generator functions with yield
- Generator expressions
- Decorators — basics & use cases
- List, dict & set comprehensions (deep)
- Walrus operator :=
- Unpacking with *
- String formatting: f-strings & format()
- Python best practices & PEP 8 style

## HANDS-ON PROJECTS

Apply everything you learn — build real programs that go in your portfolio.

PROJECT 01

### WhatsApp Chat Analyser

Parse and analyse a real WhatsApp chat export using Python. Uncover message counts, most active hours, top senders, emoji usage, and word frequency — then generate a clean summary report saved to a file.

**What you'll apply:**

- ✓ File I/O — reading .txt WhatsApp exports
- ✓ String parsing & RegEx for message extraction
- ✓ Dictionary & Counter for frequency analysis
- ✓ Datetime parsing & time-based grouping
- ✓ f-strings for formatted summary reports

PROJECT 02

### Personal Expense Tracker

Create a fully functional expense tracking application with categories, monthly summaries, and data persistence. Uses CSV files for storage and provides filtered reports on demand.

**What you'll apply:**

- ✓ OOP: Expense & Tracker classes
- ✓ CSV read/write with the csv module
- ✓ Data filtering & aggregation
- ✓ Exception handling for file errors
- ✓ Menu-driven command-line interface

## WHAT YOU WILL KNOW AFTER THIS COURSE

By Week 6 — you will confidently do all of these:

- ✓ Write clean, readable Python from scratch
- ✓ Use all data structures (list, dict, set, tuple) like a pro
- ✓ Apply OOP principles to build real programs
- ✓ Handle exceptions and errors gracefully
- ✓ Read, write & process files (txt, CSV, JSON)
- ✓ Build modular, reusable code with functions & classes
- ✓ Use Python's built-in library & install packages with pip
- ✓ Solve common coding interview problems in Python
- ✓ Build complete command-line applications
- ✓ Confidently start learning NumPy, Pandas & Data Analytics

## WHO IS THIS FOR?

- **Fresh Graduates**
  BCA, B.Tech, BSc, MBA, BBA — any stream
- **Complete Beginners**
  Zero coding experience? Perfect starting point
- **Working Professionals**
  Add Python to your skillset quickly
- **Aspiring Data Scientists**
  The essential first step before ML & AI
- **Entrepreneurs**
  Automate tasks & build your own data tools

## WHAT COMES NEXT?

- **Data Analytics**
  NumPy, Pandas, Matplotlib, Seaborn
- **Machine Learning**
  Scikit-Learn, model building & evaluation
- **Deep Learning**
  TensorFlow, Keras, CNNs & LSTMs
- **Generative AI**
  ChatGPT API, LangChain, RAG pipelines
- **Full Data Scientist**
  Complete 8-month program pathway

# Start Your Python Journey Today!

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